Time Series and Dynamics Econometrics

Assignment 3

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Part I: Spurious Regression and Unit Roots

Question 1

We simulate independent random walks:

$$X_t = X_{t-1} + v_t, \quad v_t \sim NID(0,1)$$
$$Y_t = Y_{t-1} - w_t, \quad w_t \sim NID(0,1)$$

and perform a static regression Y_t on X_t by:

$$Y_t = \alpha + \beta X_t + \varepsilon_t$$

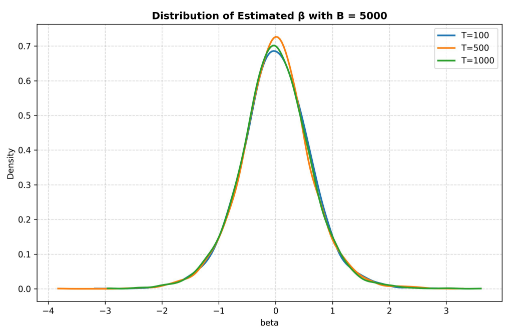

Figure 1.1 Distribution of estimated β with different sample size T

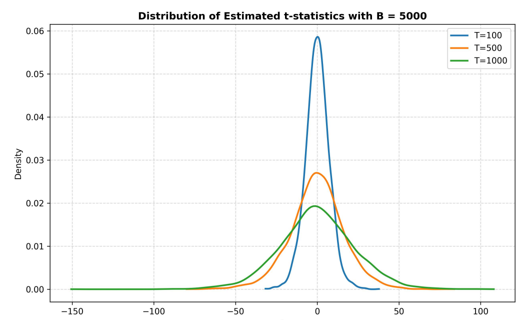

Figure 1.2 Distribution of t-statistics with different sample size T

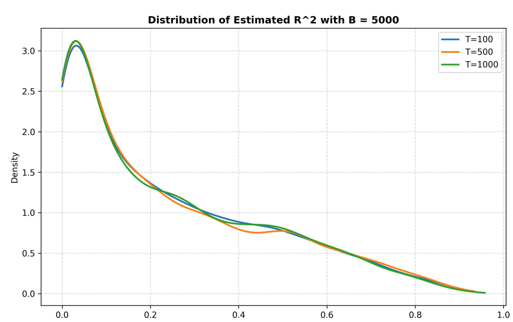

Figure 1.3 Distribution of estimated R^2 with different sample size T

In Figure 1.1 and 1.2 reveal the plots of the simulated distributions show that as the sample size T increases, the estimated β does not converge to 0. This occurs because both X_t and Y_t are non-stationary I(1) processes whose stochastic trends dominate their correlation, leading to non-standard asymptotic behavior. The distribution of t-statistics diverges instead of converges to a standard normal distribution $N(0,1)$.

Additionally, Figure 1.3 also reveals that R^2 remains large, which indicates that Y_t is highly positively correlated to X_t. The estimated coefficients often appear statistically significant with large R^2. Standard theory does not apply and statistical results can be misleading. It shows that the spurious regression problem, meaning trending non-stationary variables can create false evidence of correlation.

Question 2

In Figure 2.1 and 2.2 show the daily prices of APPLE and MICROSOFT from from the 14th of February 2007 to the 28th of January of 2013. Both series show strong upward trends and persistent fluctuations. Figure 2.3 and 2.4 illustrate the 12-period ACFs of both time series are highly positive and decay slowly, implying strong persistence and potential unit root pattern. The plots suggest that stock prices are likely $I(1)$ processes rather than stationary.

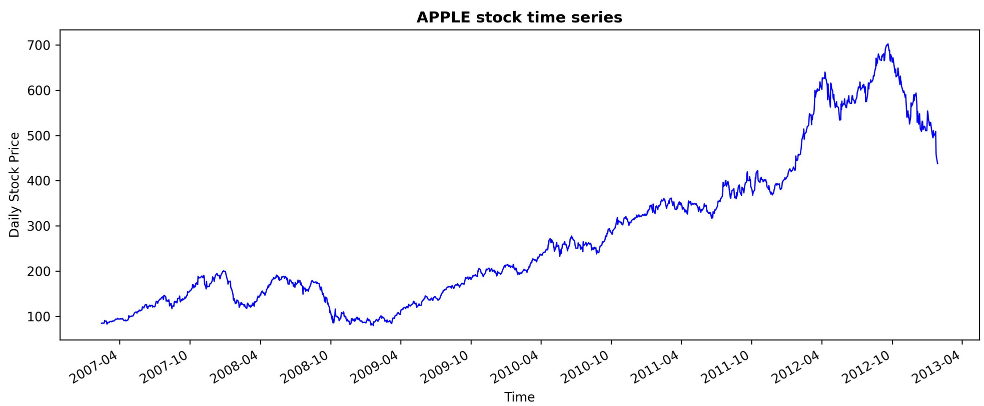

Figure 2.1 Daily stock prices of APPLE from 2007 to 2013

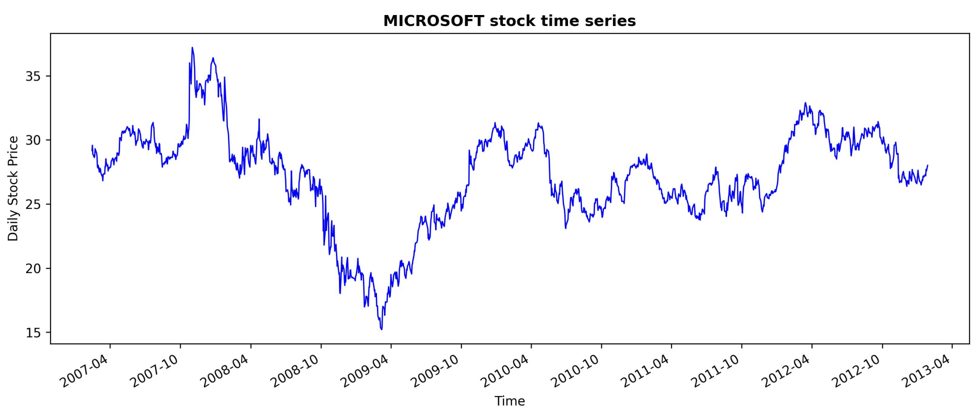

Figure 2.2 Daily stock prices of MICROSOFT from 2007 to 2013

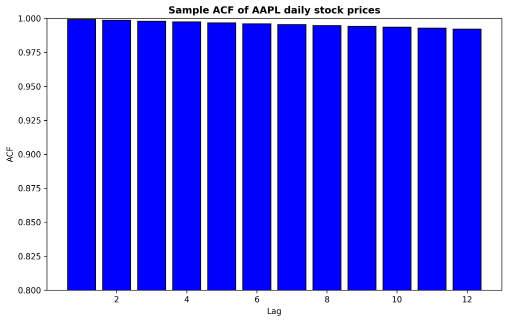

Figure 2.3 The Sample autocorrelation function of APPLE up to 12 lags

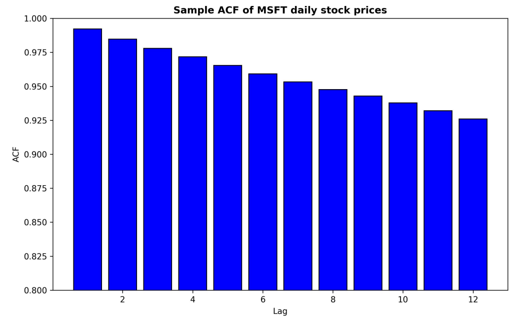

Figure 2.4 The Sample autocorrelation function of MICROSOFT up to 12 lags

Question 3

Using the Bayesian Information Criterion to choose the lag order p, we estimate simplified Augmented Dickey-Fuller (ADF) regression. For the best ADF specification based on the BIC, then compute the ADF test statistics for ADF test. Since there is no intercept, use the DF distribution, which has a critical value -1.62 at 10% significance level. Every ADF statistics are above the 10% critical values, so we cannot reject the unit root hypothesis for any stock price series. And this matches with the financial theory that the stock prices follow a random walk process. This result consistent with the Efficient Market Hypothesis, which implies that stock prices follow a random walk and are therefore non-stationary.

Stock ID	Order p (by BIC)	ADF Statistic	Decision
APPLE	2	0.892064	Do not reject
EXXON_MOBIL	3	0.174943	Do not reject
FORD	1	0.064445	Do not reject
GEN_ELECTRIC	2	-1.340003	Do not reject
INTEL	1	-0.340715	Do not reject
MICROSOFT	1	-0.412781	Do not reject
NETFLIX	1	0.143069	Do not reject
NOKIA	1	-1.218262	Do not reject
SP500	2	-0.143941	Do not reject
YAHOO	2	-1.237010	Do not reject

Table 3.1: The results of order p and Augmented Dickey-Fuller (ADF) unit root test results for each stock

Question 4

We estimate the stock prices of Microsoft and Exxonmobil has below relation:
$P_{Microsoft} = \alpha + \beta * P_{Exxonmobil} + \epsilon_t$

The estimates slope β is statistically significant and R^2 is relatively high, the relationship is spurious as both stock prices are non-stationary. The apparent significance increases from common trending behavior rather than a true economic relationship. Therefore, price changes in Microsoft stock cannot determined to be impacted by the fluctuations of Exxon Mobil stock prices.

JB Test Statistic	Critical Value (5% at χ^2_2)
27.33	5.991

Table 4.1 Jarque-Bera (JB) test statistic and the critical value of χ^2_2 distribution under the null

Using the Breusch-Godfrey test to test for autocorrelation up to lag k of the residuals of the AR(p) model, the white noise assumption for ε_t can be checked. Based on the result shown in Table 4.2, since the BG

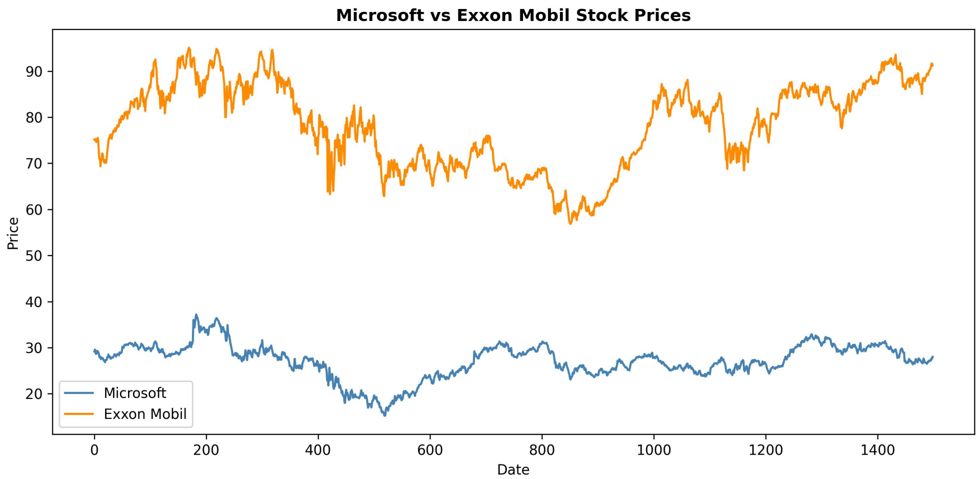

Figure 4.1 The stock price of MICROSOFT vs EXXON MOBIL

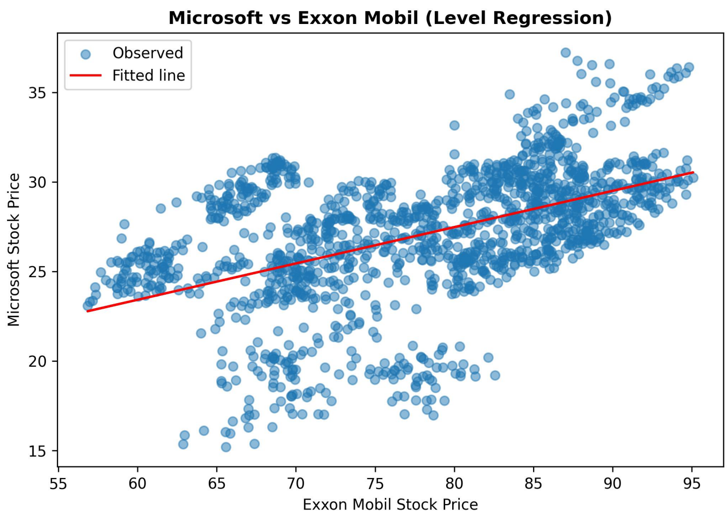

Figure 4.2 The scatter plot of stock prices of MICROSOFT vs EXXON MOBIL

test statistic is smaller than the 95% critical value of the χ^2_4 under the null, therefore do not reject the null hypothesis as the residuals has no autocorrelation.

BG Test Statistic	Critical Value (5% at χ^2_4)
0.0	9.488

Table 4.2 Breusch-Godfrey (BG) test statistic and the critical value of χ^2_4 distribution under the null

Part II: Cointegration and Error Correction Models

Question 1

In Figure 5.1, 5.2 and 5.3 reveal the plots of the simulated distributions of the regression $Y_t = \alpha + \beta X_t + \varepsilon_t$ for different sample sizes T when two $I(1)$ time series $\{Y_t\}$ and $\{X_t\}$ are cointegrated. Figure 5.1 shows that the estimated β converge to 0 as the sample size T increases. Figure 5.2 illustrates the distribution of t-statistics converges to a standard normal distribution $N(0,1)$. Finally, Figure 5.3 suggests that R^2 successfully reflects the true correlation as the sample size T increases, which indicates that Y_t is highly positively correlated to X_t. Standard theory applies. It shows that the spurious regression problem does not occur when two $I(1)$ time series are cointegrated.

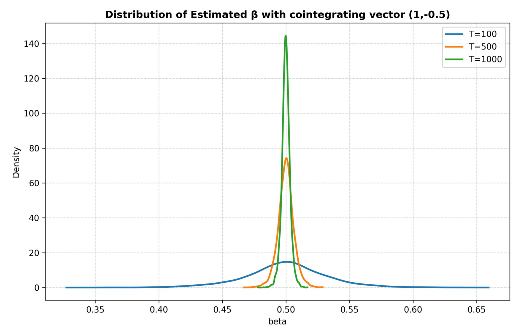

Figure 5.1 Distribution of estimated β with different sample size T

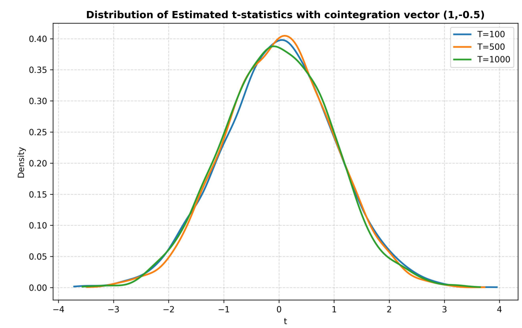

Figure 5.2 Distribution of t-statistics with different sample size T

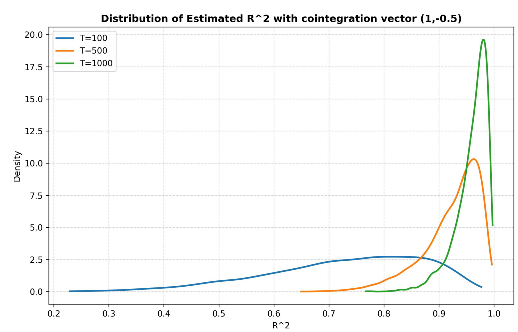

Figure 5.3 Distribution of estimated R^2 with different sample size T

Figure 5.4 and 5.5 compare the outcomes of the same regression $X_t = \phi X_{t-1} + v_t$, where $v_t \sim N(0,1)$, with X_t generated as $\phi = 0.5$ and $\phi = 0.25$ respectively. Both of the results are obtained in some very similar distributions. It supports the super-consistency of the estimator of the static cointegrating regression.

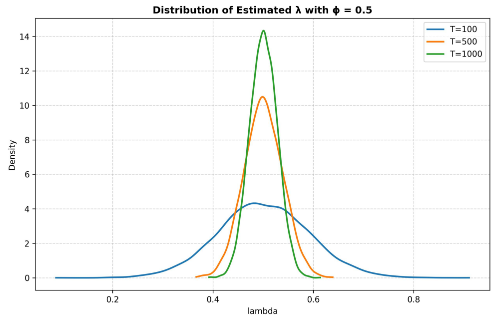

Figure 5.4 Distribution of estimated λ with different sample size T for $\phi = 0.5$

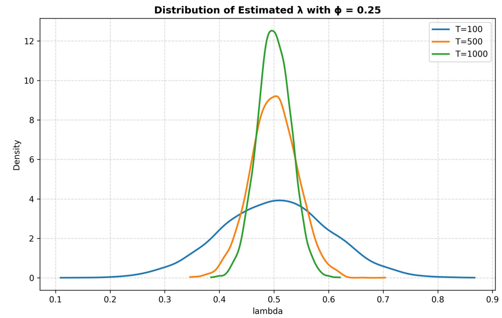

Figure 5.5 Distribution of estimated λ with different sample size T for $\phi = 0.25$

Question 2

Figure 6.1 plots both the quarterly aggregate consumption and aggregate income of families in the Netherlands. Both series show gradual upward trends while there is a slightly decrease in 2008. Figure 6.2 and 6.3 illustrate the 12-period ACFs of both time series are highly positive and decay slowly, implying strong persistence.

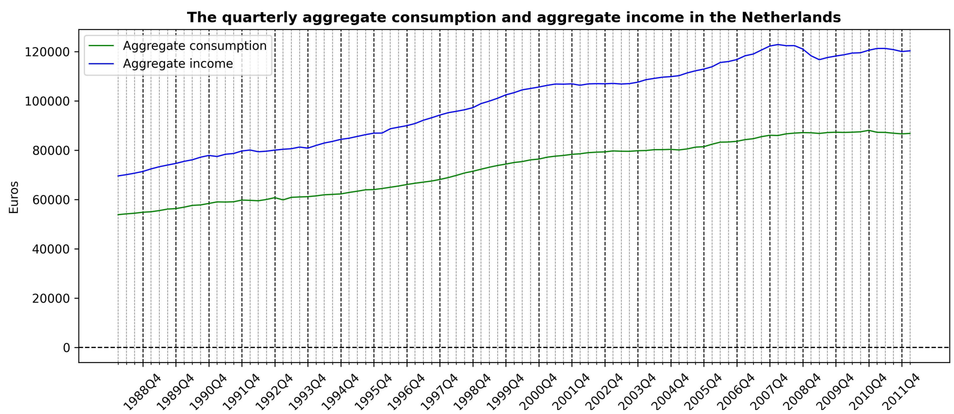

Figure 6.1 Quarterly aggregate consumption and income in the Netherlands

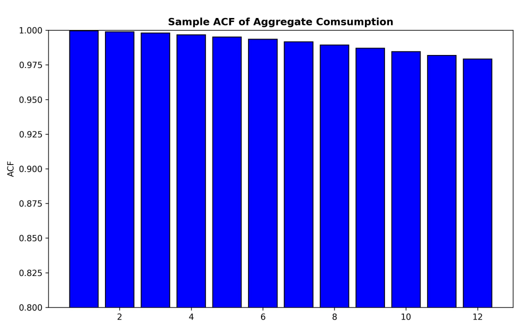

Figure 6.2 The SACF of quarterly aggregate consumption up to 12 lags

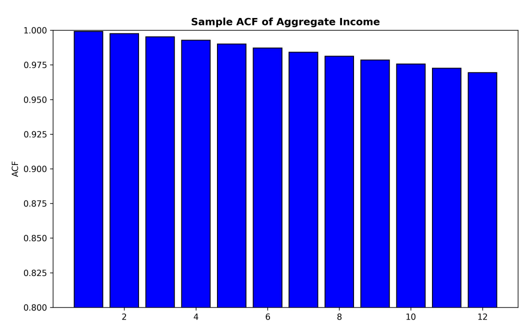

Figure 6.3 The SACF of quarterly aggregate income up to 12 lags

Question 3

We first estimate the static long-run relationship between aggregate consumption (Y_t) and disposable income (X_t) using the regression:

$$Y_t = \alpha + \beta X_t + Z_t$$

The estimated coefficients $\hat{\alpha}$ and $\hat{\beta}$ would be calculated. The estimated regression coefficients are:

$$\hat{\alpha} = 6783.373 \quad \text{and} \quad \hat{\beta} = 0.665$$

Next, obtain the estimate residuals $\hat{Z}_t$ using the estimated coefficients $\hat{\alpha}$ and $\hat{\beta}$ in the estimated $\hat{Y}_t$ as

$$\hat{Z}_t = Y_t - \hat{\alpha} - \beta \hat{X}_t \approx Z_t$$

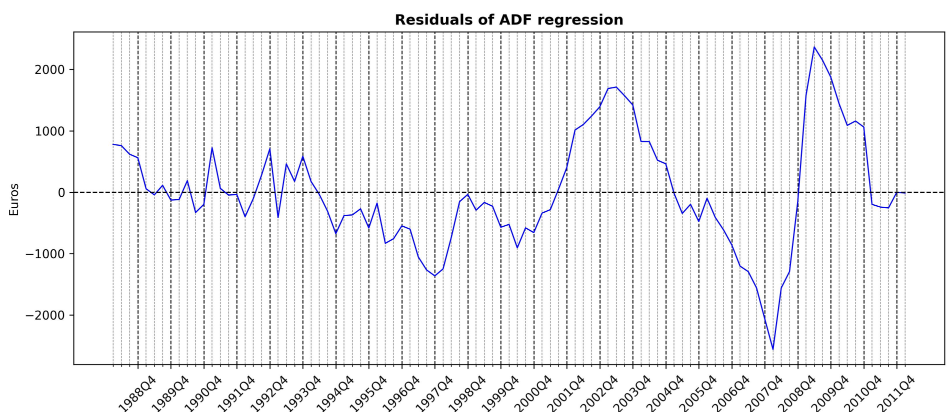

Figure 7.1 The residuals of ADF regression by regressing consumption on income

Figure 7.1 reveals the plot of the regression residuals. Furthermore, to test the cointegration between Y_t and $\{X_t\}$, perform a unit root test in the residuals to determine the integration order of Z_t. Use the Akaike Information Criterion (AIC) to determine the number of ADF lags in the ADF regression in the unit root residual test. The test hypothesis of the unit root residual test is:

$$H_0 : \beta = 0 \Rightarrow \{Z_t\} \sim I(1) : \text{No-cointegration} \quad \text{and} \quad H_1 : \beta < 0 \Rightarrow \{Z_t\} \sim I(0) : \text{Cointegration}$$

Since the ADF test is left-sided, reject the null for small values of ADF statistic. There is no intercept in ADF-Cointegration test test, therefore the critical value $ADFNC_{crit}$ at the 10% significant level is approximately -1.62. The cointegration test statistic is

$$ADFNC_{calc} = -3.428$$

which is smaller than the $ADFNC_{crit} = -1.62$. Since $ADFNC_{calc} < ADFNC_{crit}$, reject the null hypothesis of no-cointegration, and accept the alternative which supports cointegration by the evidence $\{\hat{Z}_t\} \sim I(0)$. As a result, the ADF-Cointegration Test leads to means that quarterly aggregate *consumption* and *income* are conintegrated.

Question 4

Since $(Y_t, X_t) \sim CI(1, 1)$, the Granger's Representation Theorem applies. $\{Y_t\}$, which is the time series of quarterly aggregate *consumption*, admits the error-correction representation:

$$\Delta Y_t = \gamma[Y_{t-1} - \delta - \lambda X_{t-1}] + \sum_{i=1}^{p} \phi_i \Delta Y_{t-i} + \sum_{i=1}^{q} \beta_i \Delta X_{t-i} + \varepsilon_t$$

where $\{X_t\}$, which is the time series of quarterly aggregate *income*. Using the estimated residuals from the cointegration regression $\hat{Z}_t$, denote $\hat{Z}_{t-1} = Y_{t-1} - \hat{\delta} - \hat{\lambda} X_{t-1}$, estimate an Error Correction Model (ECM) for consumption growth, the function as follows:

$$\Delta Y_t = \gamma \hat{Z}_{t-1} + \sum_{i=1}^{p} \phi_i \Delta Y_{t-i} + \sum_{i=1}^{q} \beta_i \Delta X_{t-i} + u_t$$

where u_t is the residual from the cointegration regression, representing the deviation from the long-run equilibrium. Again, use the AIC to select the model specification (p, q). We select the lag orders (4,6). The estimated error-correction model is:

$$\begin{aligned}\Delta Y_t = &- 0.051\hat{Z}_{t-1} - 0.118\phi_1 \Delta Y_{t-1} + 0.134\phi_2 \Delta Y_{t-2} + 0.506\Delta Y_{t-3} + 0.172\Delta Y_{t-4} \\ &+ 0.154\Delta X_{t-1} - 0.068\Delta X_{t-2} + 0.041\Delta X_{t-3} - 0.025\Delta X_{t-4} + 0.058\Delta X_{t-5} - 0.048\Delta X_{t-6} + u_t\end{aligned}$$

The long-run equilibrium in the error-correction model is:

$$0 = \gamma[\bar{Y} - \hat{\delta} - \hat{\lambda}\bar{X}] + 0 + 0 + 0 \Rightarrow \bar{Y} = 6783.373 + 0.665\bar{X}$$

Based on the estimated error-correction model above, the value of $\gamma = -0.051$. Since $\gamma < 0$, it suggests that correction of Y_t towards long-run equilibrium. In other words, γ would corrects the *consumption* adjust to restore equilibrium after short-run shocks.